

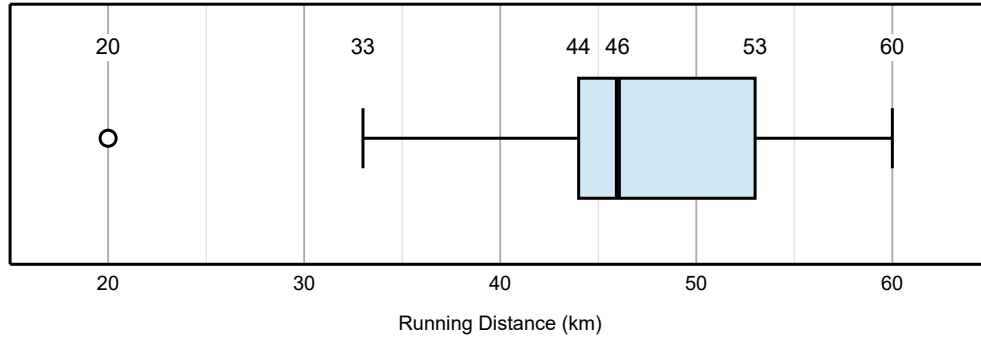
Introduction to Statistics - Quiz #1 (60 minutes)

September 17, 2026 (Thursday)

Section(교반): _____ Cadet Number(교번): _____ Name(성명): _____ Score: _____

- All solutions (except for the first problem) must include a detailed step-by-step explanation.
- If an answer has more than four decimal places, round to the fourth decimal place or write it as a fraction.

1. We randomly selected 17 students at a university and recorded each student's running distance (km) during the month of May. The boxplot below shows the results. Fill in each blank with the correct value. [30 points]



- (1) The median of the running distance is (_____).
- (2) The first quartile Q_1 of the running distance is (_____).
- (3) The outlier of the running distance is (_____).
- (4) The interquartile range (IQR) of the running distance is (_____).
- (5) About (_____)% of all the data values are between 44 and 53.

2. We surveyed 100 households in an apartment complex about the number of TVs X and the number of cars Y they have. The table below shows the number of households. Answer the following questions. [70 points]

$y \backslash x$	1	2	3	4
0	20	15	10	0
1	10	20	7	3
2	4	5	4	2

- (1) Fill in the table below with the joint probability distribution $f(x, y)$ of the random variables X and Y , and the marginal probability mass functions $f_X(x)$ and $f_Y(y)$. To find $f(x, y)$, divide each number in the table above by the total number of households, 100. For example, 20 households have 1 TV and 0 cars, so $f(1, 0) = 20/100 = 0.20$.

$y \backslash x$	1	2	3	4	$f_Y(y)$
0	0.20				
1					
2					
$f_X(x)$					

- (2) Calculate $E(XY)$, $E(X)$, and $E(Y)$.

- (3) Calculate $\text{Cov}(X, Y)$.

- (4) Determine whether the random variables X and Y are independent.

- (5) One household is randomly selected. Let A be the event that this household has at least one car: $Y \geq 1$. Find $P(A)$.

- (6) One household is randomly selected. Let B be the event that this household has three or more TVs: $X \geq 3$. Find $P(B \mid A)$.